Supervised Regression and the Bias–Variance Trade-off

Machine Learning, Homework 2

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Notebook (`supervised_regression.ipynb`). I described how scikit-learn maps the hypothesis space, risk and optimisation (HRO) view of learning onto estimators, `fit` and scoring. I listed the attributes of the Iris dataset object and the hyperparameters of `DecisionTreeRegressor`. I also wrote the empirical risk and gradient-descent update for a cubic polynomial model. On the UCI Abalone data I predicted the number of rings from the longest shell length and the whole weight with linear regression (70/30 split). The fit is $\hat{y} = 3.54 + 10.33\,\text{length} + 1.18\,\text{weight}$, with test MAE 1.95 rings (Figure 1).

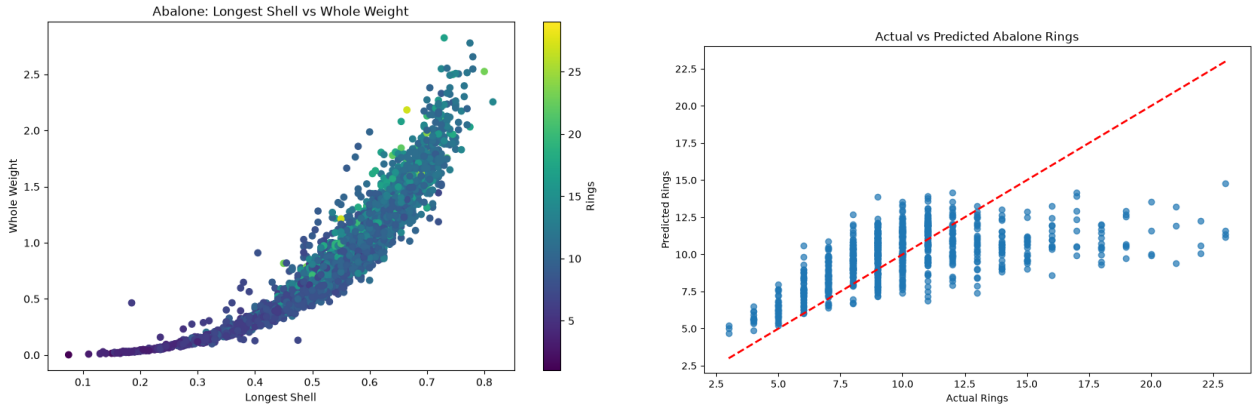


Figure 1: Abalone: inputs coloured by rings (left), predicted vs. observed rings on the test set (right).

Written part.

Least-squares line (`Q2_a.py`, `Q2_b.py`). For the commute data, $y = 18.50 + 0.564x$. If the teacher leaves x minutes after 7:00 and must arrive by 8:40, the number of minutes early is $z = (100 - a) - (1 + b)x$. Setting $z = 0$ gives $x \approx 52$, so the latest departure is about 7:52.

Tree age from height and circumference. With a linear model and only two inputs, the bias is high: tree type, climate and location matter and the relation is not linear. The variance is low, because the model is simple and has a million samples. The irreducible error comes from trees with the same size but different ages. The largest gain is in reducing bias, for example with more inputs and non-linear features.

Bias–variance decomposition. I showed $\mathbb{E}_{D,\varepsilon}[(y - \hat{y}_D)^2] = \text{Bias}^2 + \text{Var} + \sigma^2$ and explained the trade-off with K in KNN (small K: high variance, large K: high bias). Bagging averages models, so it lowers variance and can raise bias slightly. Boosting fits the residuals sequentially, so it lowers bias and can raise variance.

Ridge regression as MAP. With prior $w \sim \mathcal{N}(0, \alpha^2 I)$ and Gaussian noise of variance σ^2, the MAP estimate equals the ridge solution with $\lambda = \sigma^2/\alpha^2$.